BANKING DATA PLATFORM MIGRATION

Synapse → Databricks Migration Proposal

A phased, risk-managed strategy for migrating 100 production pipelines from Azure Synapse to Databricks

100	4	10–16	4–6
PIPELINES IN SCOPE	APPROACHES EVALUATED	WEEKS (RECOMMENDED)	ENGINEERS

Prepared for the Data Platform Modernization Initiative

Summary

Why this proposal exists, and what it recommends

The bank's Synapse workspace contains 100 pipelines covering ingestion, transformation, validation, and reporting across core banking domains. As the organization moves its processing platform to Databricks, business logic must be preserved and downstream risk minimized.

Rather than assume a single migration style fits all 100 pipelines, this proposal evaluates four realistic approaches — **One-to-One, Consolidation/Re-engineering, Template/Framework-Based, and Hybrid** — and compares them on time, effort, feasibility, and long-term maintainability.

Recommendation: the Hybrid Migration approach, which classifies pipelines by complexity and applies the right migration strategy to each group.

CORE CHALLENGE

100 pipelines of mixed complexity, one migration deadline

APPROACH EVALUATED

4 realistic strategies, benchmarked side-by-side

RECOMMENDATION

Hybrid Migration — classify, template, customize, wave-migrate

Assumptions & Scope

The baseline every estimate in this proposal is built on

1 Platform

- 100 existing Synapse pipelines
- Mix of simple, medium, and complex pipelines
- ADLS Gen2 remains the storage layer
- Databricks is the target processing platform

2 Workload

- Pipelines span ingestion, transformation, validation, and reporting
- Existing business logic must be preserved
- Testing and production cutover are in scope

3 Delivery

- Assuming a team of roughly 4–6 engineers available
- Estimates are calendar-time, assuming parallel work

1

One-to-One Migration

Recreate every Synapse pipeline as its own Databricks job

THE IDEA Each of the 100 Synapse pipelines is analyzed individually and rebuilt as its own Databricks job — Customer Pipeline → Customer Job, Account Pipeline → Account Job, and so on. Architecture stays almost identical to today.

TIME	EFFORT	FEASIBILITY	MIGRATION RISK
12–16 weeks	High (~20–30 person-months)	Very High	Low–Medium

PROS

- ✓ Lowest architectural risk
- ✓ Straightforward to execute
- ✓ Easy to validate vs. Synapse output
- ✓ Minimal disruption to business logic

CONS

- ✗ High code duplication
- ✗ 100 workflows to maintain long-term
- ✗ Doesn't leverage Databricks capabilities
- ✗ Technical debt carries over from Synapse

USE THIS WHEN

The priority is moving off Synapse as quickly and safely as possible, without changing much.

KEY RISKS

- Duplicated code across pipelines
- Inconsistent implementations
- Higher deployment/monitoring overhead
- Underuses Databricks platform

2

Consolidation / Re-engineering

Redesign around business processes, not 1:1 pipeline mapping

THE IDEA Instead of preserving the 100-pipeline structure, common patterns are analyzed and consolidated. E.g. Customer_Load, Customer_Clean, Customer_Validate, Customer_Gold collapse into one Customer Processing Job. ~100 pipelines become ~30–50 Databricks jobs.

TIME	EFFORT	FEASIBILITY	MIGRATION RISK
16–24 weeks	**Very High (~30–45 person-months)**	**Medium**	**High**

USE THIS WHEN

The objective isn't just to move platforms — the organization wants to fundamentally modernize the data platform.

PROS

- ✓ Major reduction in duplicated code
- ✓ Better long-term architecture
- ✓ Easier centralized maintenance
- ✓ Full use of Databricks capabilities

CONS

- ✗ Highest initial effort and longest timeline
- ✗ Business logic may unintentionally change
- ✗ Difficult to test comprehensively
- ✗ Can balloon into an open-ended redesign

KEY RISKS

- Scope creep beyond migration
- Complex dependencies missed
- Large up-front design effort
- Hard to compare old vs. new behavior

3 Template / Framework-Based

Build once, configure many — for pipelines that share a shape

THE IDEA Roughly 60 of 100 pipelines follow the same Ingestion → Silver → Validation → Gold pattern. A reusable framework is built once; individual pipelines then supply configuration and business rules rather than full custom code.

USE THIS WHEN

Many pipelines share a similar pattern: Source → Clean → Validate → Transform → Gold.

TIME	EFFORT	FEASIBILITY	MIGRATION RISK
10–14 weeks	**Medium–High (~18–28 person-months)**	**High***	**Medium**

PROS

- ✓ High reuse, less duplicated code
- ✓ Faster migration after framework is built
- ✓ Consistent implementation across pipelines
- ✓ Good scalability

CONS

- ✗ Not suitable for every pipeline
- ✗ Complex pipelines need exceptions
- ✗ Framework requires careful upfront design
- ✗ Risk of over-engineering

KEY RISKS

- Framework becomes too generic
- Unique business rules don't fit
- *Feasible only if patterns are common
- Initial design takes longer than planned

4 Hybrid Migration ⭐ Recommended

Classify first, then match strategy to complexity

THE IDEA The 100 pipelines are classified by complexity: ~40 simple pipelines use a reusable template, ~45 medium pipelines use the template plus customization, and ~15 complex pipelines (e.g. fraud detection) get an individual redesign.

USE THIS WHEN

This is the recommended approach for the 100-pipeline banking Synapse workspace.

TIME	EFFORT	FEASIBILITY	MIGRATION RISK
10–16 weeks	Medium–High (~20–30 person-months)	Very High	Low–Medium

PROS

- ✓ Best balance of speed, quality, and risk
- ✓ Reuse where it makes sense, custom where needed
- ✓ Doesn't force complex pipelines into templates
- ✓ Supports phased, wave-based migration

CONS

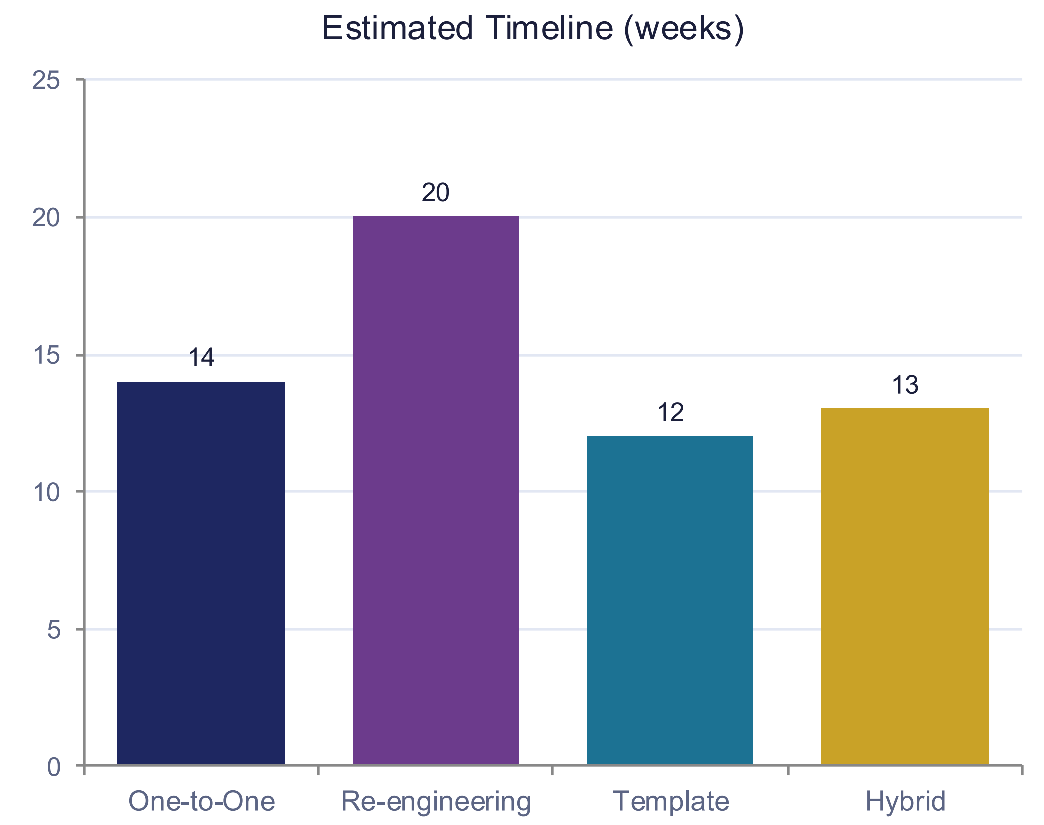
- ✗ Requires an upfront assessment phase
- ✗ Some pipelines still need individual work
- ✗ Framework development adds initial effort
- ✗ Needs careful migration governance

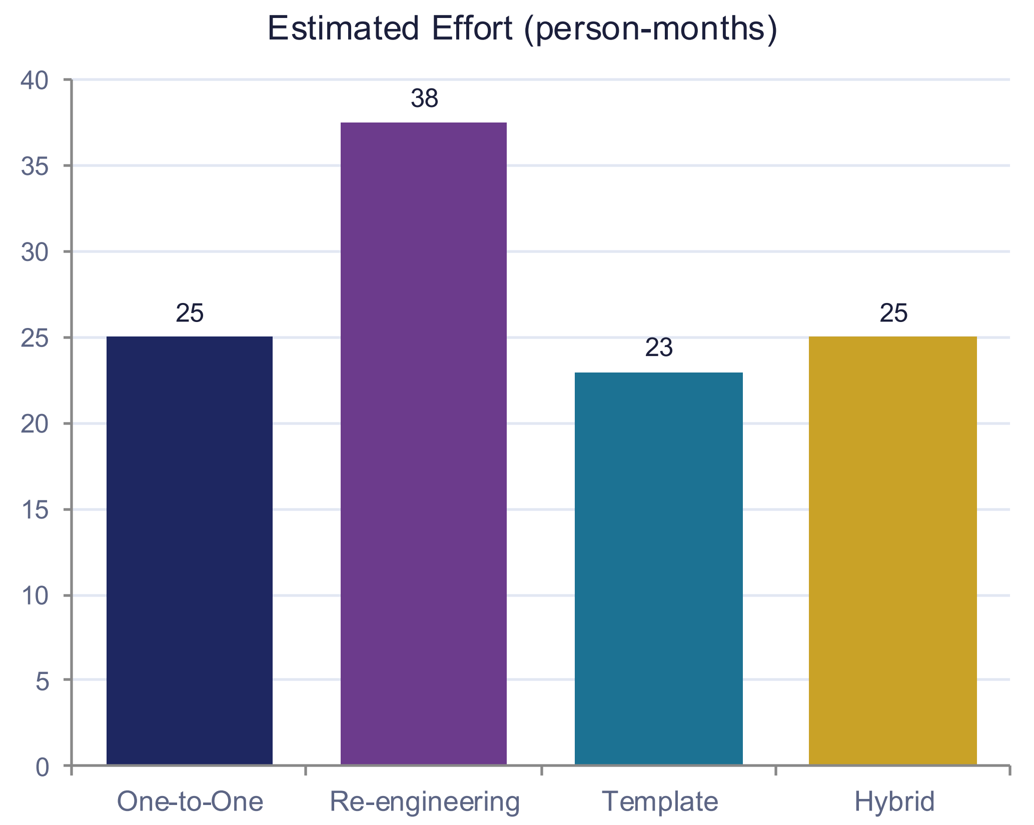
KEY RISKS

- Pipeline misclassification
- Framework doesn't cover every case
- Complex dependencies missed in discovery
- Coordination across parallel waves

Time & Effort Comparison

Midpoint estimates across the four approaches





Hybrid delivers Template-level speed while covering the full complexity range of a real enterprise workspace.

Full Comparison Matrix

All ten evaluation criteria, side by side

Criteria	One-to-One	Re-engineering	Template	Hybrid ⭐
Pipelines	100 → ~100	100 → ~30–50	100 → framework-based	100 → mixed
Time	12–16 weeks	16–24 weeks	10–14 weeks	10–16 weeks
Effort	High	Very High	Medium–High	Medium–High
Feasibility	Very High	Medium	High	Very High
Migration risk	Low–Medium	High	Medium	Low–Medium
Reusability	Low	Very High	Very High	High
Long-term maintenance	Poor	Excellent	Excellent	Very Good
Initial complexity	Low	Very High	High	Medium
Business logic preservation	Excellent	Medium	High	High
Databricks optimization	Low	Excellent	High	High
Scalability	Medium	High	Very High	Very High

Recommended Migration Roadmap

Hybrid Migration, executed in six stages

1

Inventory & Assess

Classify all 100 pipelines by complexity and dependencies

2

Pilot Migration

Migrate ~5 representative pipelines to prove the approach

3

Build Framework

Create the reusable ingestion → silver → validate → gold pattern

4

Migrate in Waves

Move simple, medium, then complex pipelines in controlled batches

5

Parallel Testing

Run Synapse and Databricks side by side to validate output

6

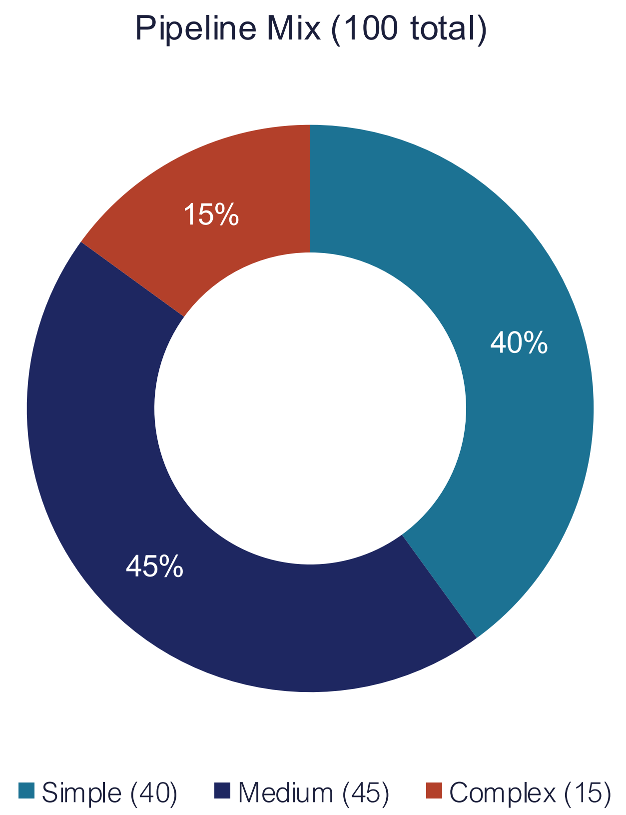
Production Cutover

Retire Synapse pipelines once validation is complete

Synapse and Databricks run in parallel during validation to control cutover risk at every wave.

Pipeline Classification

How the 100 pipelines split, and the strategy applied to each group



40

Simple

Reusable Template

Standard ingestion → silver → validate → gold pattern; migrated via configuration only.

45

Medium

Template + Customization

Framework pattern plus domain-specific business rules layered on top.

15

Complex

Individual Redesign

E.g. fraud detection — analyzed and rebuilt as custom Databricks implementations.

Why the Hybrid Approach Wins

Five reasons it fits a real enterprise banking migration

1	**It's realistic**	An enterprise's 100 pipelines will almost certainly have different levels of complexity — treating them as identical is the false assumption other approaches make.
2	**It reuses proven patterns**	The ingestion → silver → validation → gold pattern becomes a reusable processing template, not something rebuilt manually for every pipeline.
3	**It avoids over-engineering**	No attempt is made to build one giant generic framework that handles every possible banking workload.
4	**It controls migration risk**	A small pilot is migrated and validated first, before the approach is scaled to the remaining 95 pipelines.
5	**It gives a realistic timeline**	For planning purposes, ~10–16 weeks with a 4–6 person team, subject to discovery and complexity found along the way.

Recommended Approach

Hybrid Migration

For a 100-pipeline Synapse environment, a hybrid migration approach is recommended. Pipelines are first inventoried and classified by complexity, dependencies, transformation logic, and common processing patterns. Simple and repetitive pipelines migrate through reusable Databricks frameworks and parameterized workflows; medium-complexity pipelines use the same framework with pipeline-specific customization; highly complex pipelines are individually redesigned. A representative pilot runs first, followed by migration in controlled waves with parallel validation against Synapse.

Don't migrate 100 pipelines 100 different times.
Don't force 100 pipelines into one generic framework either.

Analyze → classify → template where possible → customize where necessary → migrate in waves.

THANK YOU